

Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

A reproducible context-residual estimand and the spatial scale of its recovery

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The paradox that defines premalignancy

concept schematic

a lesion field —
many indolent, one progressing

Same mutations.

Same histology.

Different outcomes.

Two explanations

Cell-intrinsic

the tissue is a bystander

Tissue-held

the tissue is a participant

**Holding a cell's own state and stage fixed,
does its tissue context change where it ends up?**

The premalignant niche: a candidate mechanism

FIGURE NOT FOUND
biorender_mechanism

The measurement will not cooperate

Profiling destroys the tissue.

No cell is ever observed twice.

So the effect cannot be measured. It has to be *estimated*.

Which means the design has to be built around falsification, not fit.

Three hypotheses, fixed before the data

- H₁** A reproducible context residual exists.
- H₂** That information is *receiver-local*.
- H₃** The estimand transfers to a second epithelium.

They do not resolve in the same direction.

That is the contribution.

Act I

The estimand is identifiable, and it is not circular

One measurement, two disjoint readings

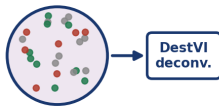
Visium spot



multicellular

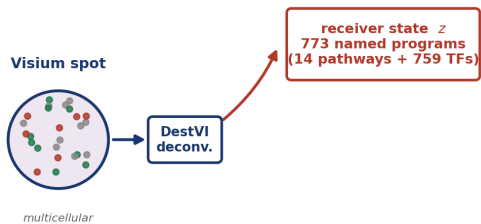
One measurement, two disjoint readings

Visium spot

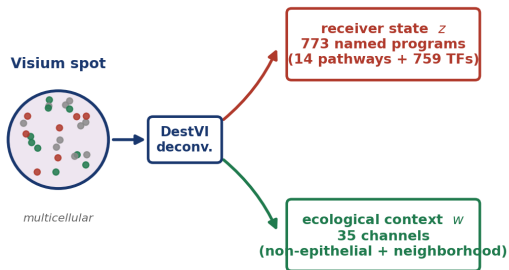


multicellular

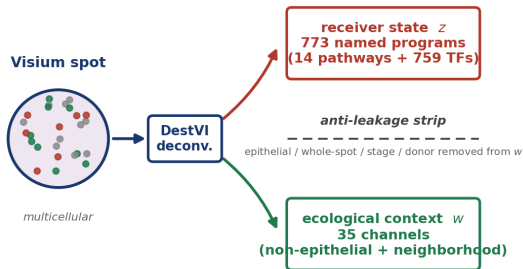
One measurement, two disjoint readings



One measurement, two disjoint readings



One measurement, two disjoint readings



the two representations are disjoint by construction

Correspondence is estimated *within* ecological strata

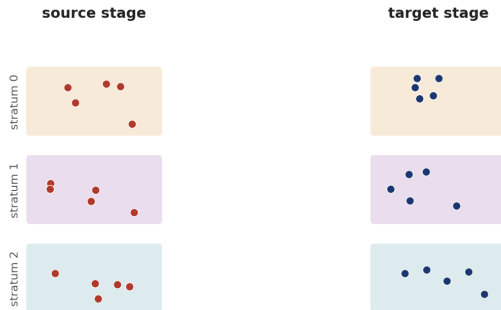
source stage



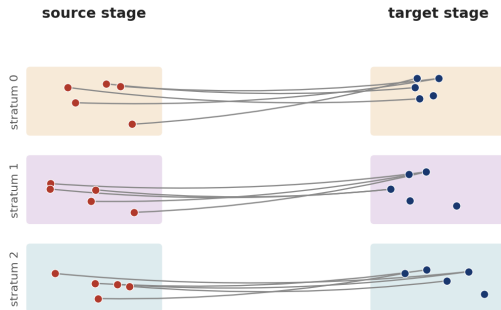
target stage



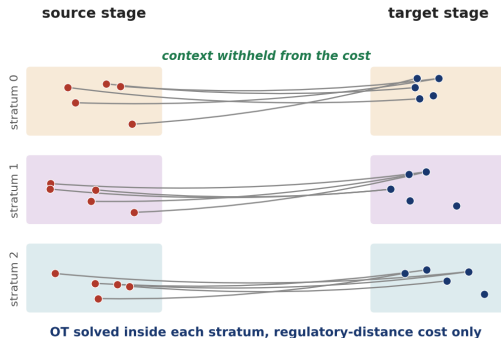
Correspondence is estimated *within* ecological strata



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Correspondence is estimated *within* ecological strata

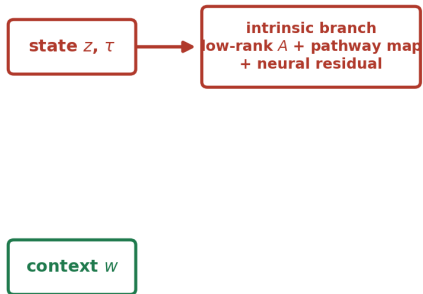


The field is structured, and context enters through a waist

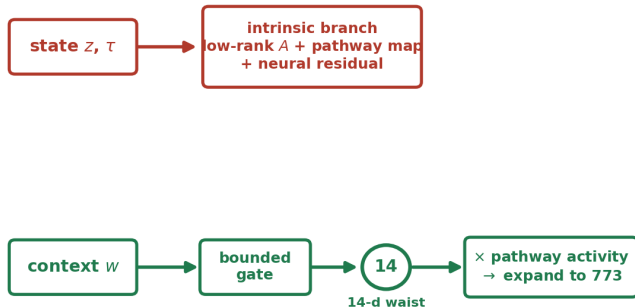
state z, τ

context w

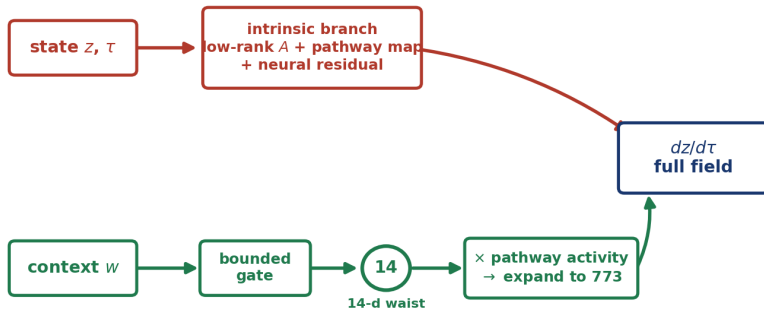
The field is structured, and context enters through a waist



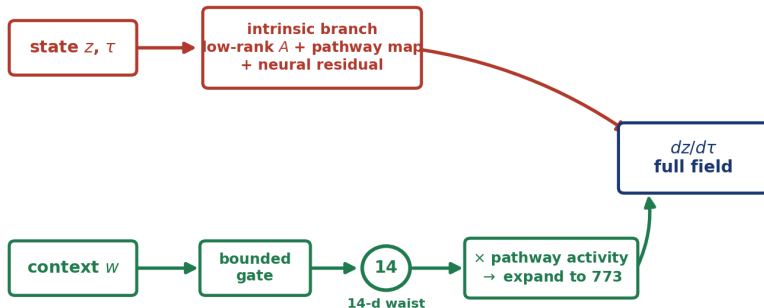
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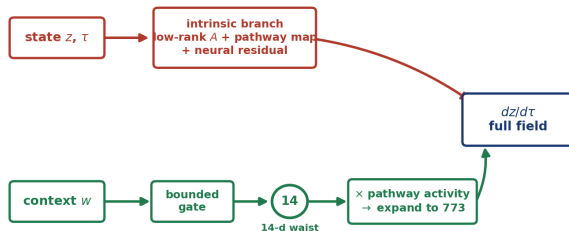


The field is structured, and context enters through a waist



759 TF coordinates move only through the 14-channel gate or the residual

The field is structured, and context enters through a waist



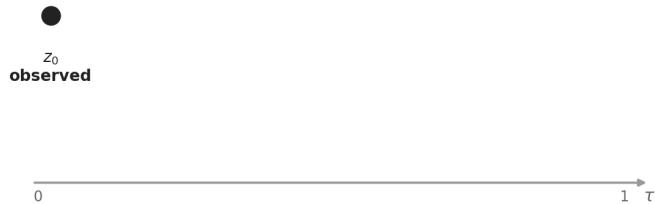
759 TF coordinates move only through the 14-channel gate or the residual

The learned field, made explicit:

$$\frac{dz}{d\tau} = \underbrace{f_{\text{self}}(z, \tau)}_{\text{intrinsic}} + \underbrace{g(w) \odot f_{\text{ctx}}(z, \tau)}_{\text{niche-gated, 14-d gate } g}$$

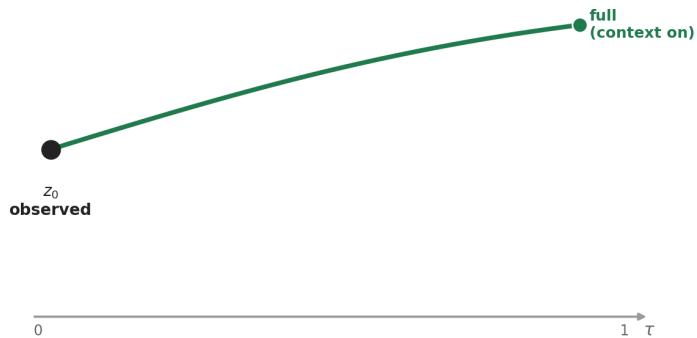
One start state, two integrations

one fit, one intrinsic field, context held fixed along both paths



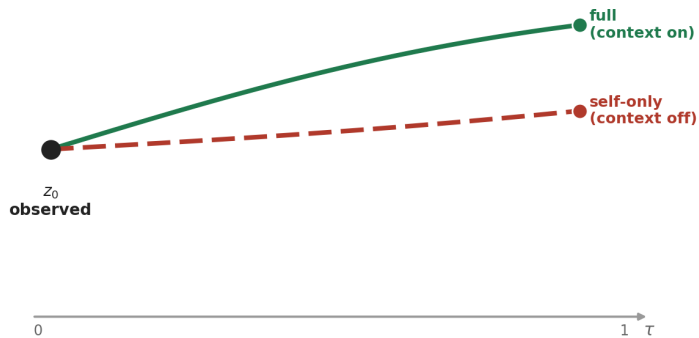
One start state, two integrations

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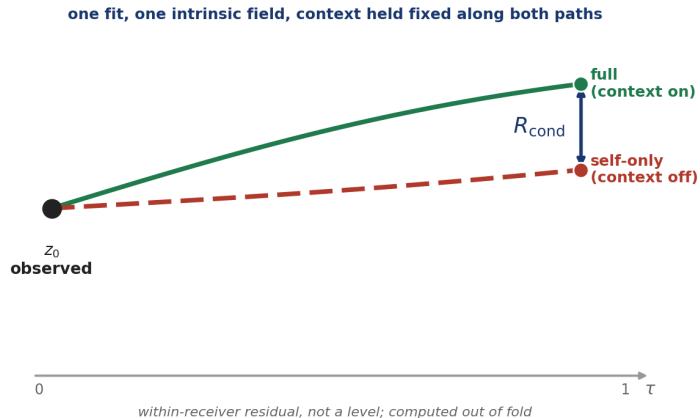


One start state, two integrations

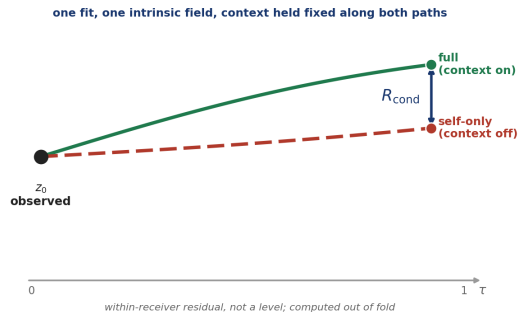
one fit, one intrinsic field, context held fixed along both paths



One start state, two integrations



One start state, two integrations



The estimand, one line:

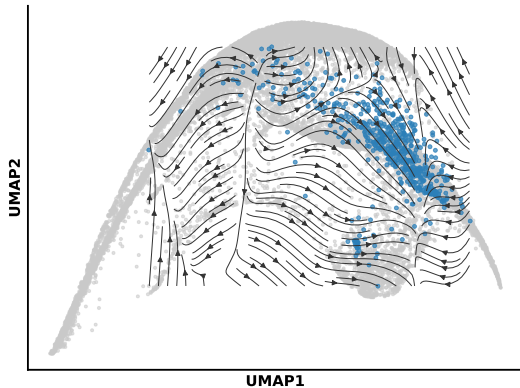
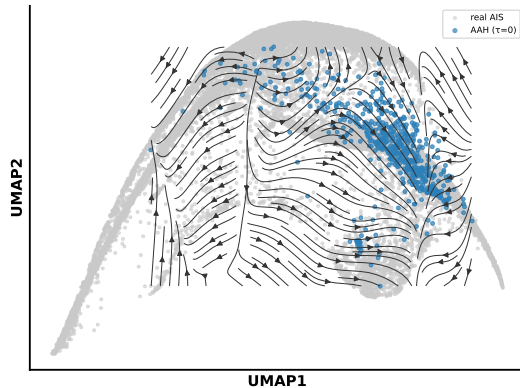
$$R_{\text{cond}}(z_i^{\text{obs}}, w_i) = z_i^{\text{full}}(1) - z_i^{\text{self}}(1) \quad (\text{same } z_0, \text{ same intrinsic field, context on vs. off})$$

The fitted field, both branches, over the real manifold

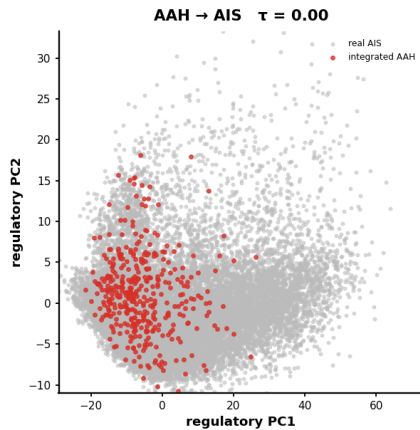
Learned velocity field: spatial context redirects AAH→AIS transport

self-only (context-neutral)

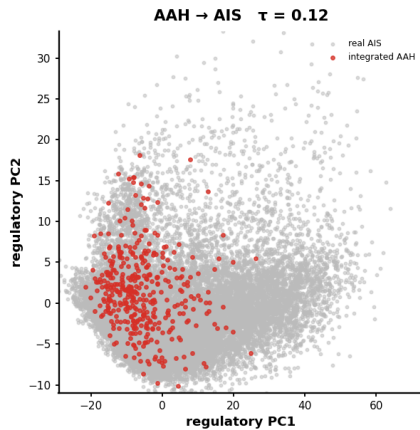
full context (niche-gated)



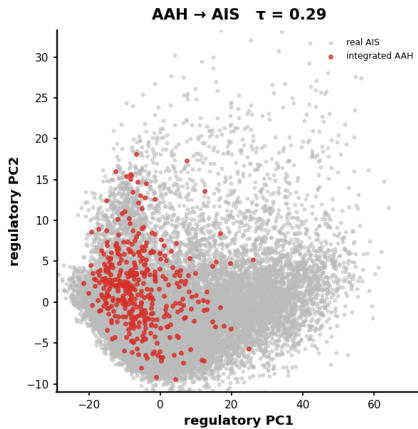
Watching the transport: integrated AAH lands on real AIS



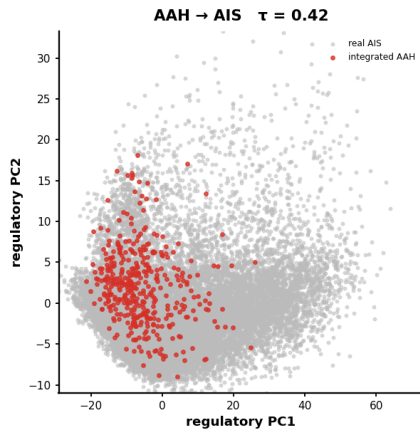
Watching the transport: integrated AAH lands on real AIS



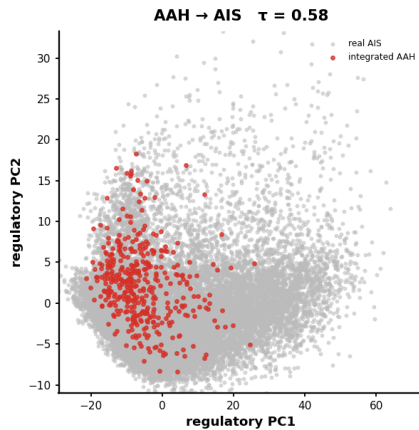
Watching the transport: integrated AAH lands on real AIS



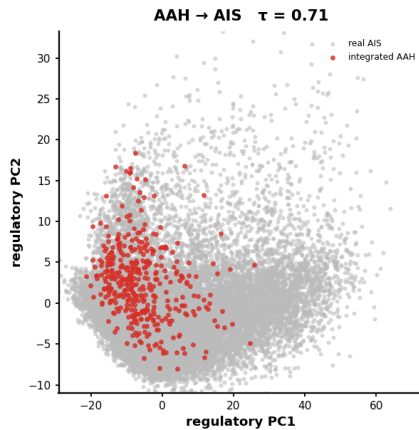
Watching the transport: integrated AAH lands on real AIS



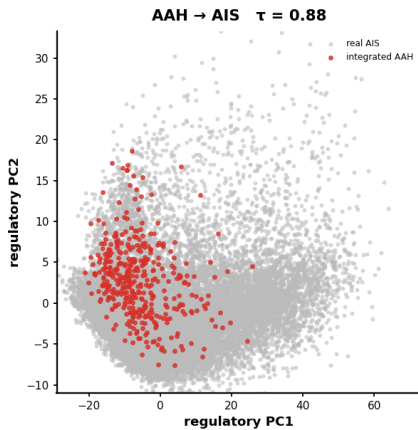
Watching the transport: integrated AAH lands on real AIS



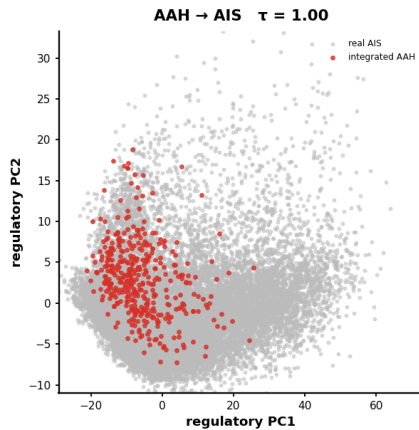
Watching the transport: integrated AAH lands on real AIS



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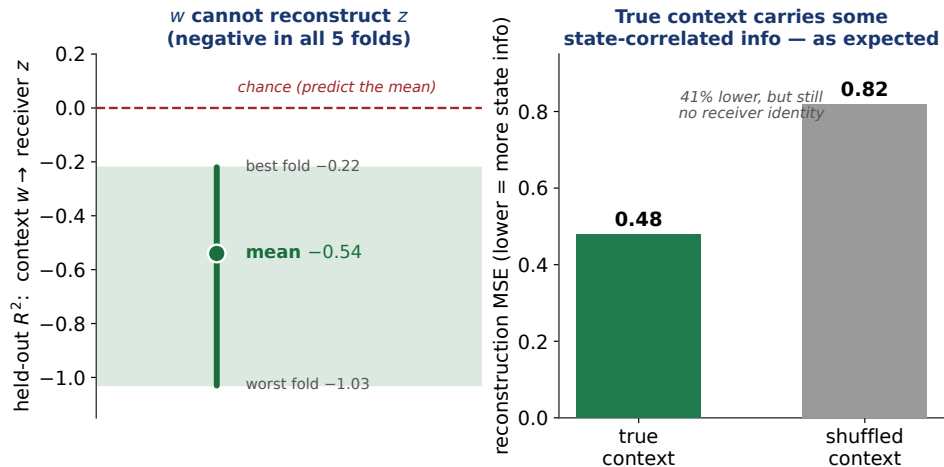


Three candidate estimands.

Two were rejected.

One of them was mine.

The context vector does not encode the receiver

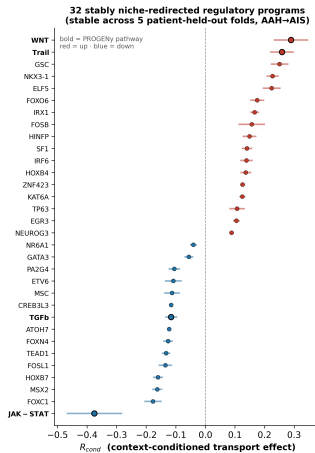


ridge regression, ecological context $\rightarrow$ receiver state, patient-held-out

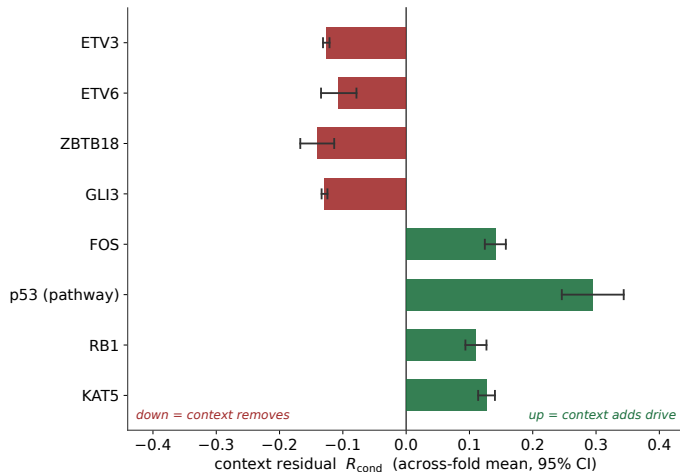
Act II

What the estimand finds

29 of 773 programs are reproducibly niche-redirected

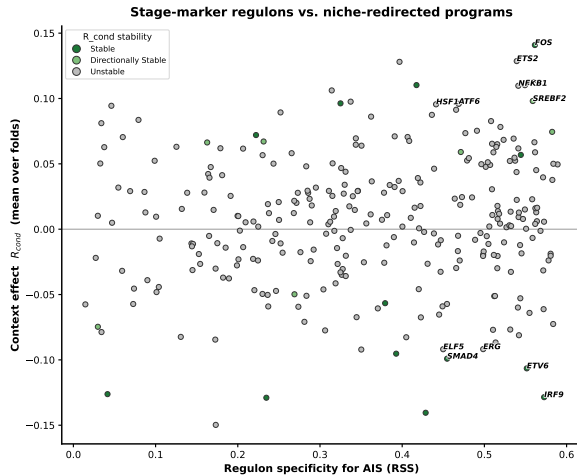


The preinvasive residual is a circuit, not a list



four repressors (ETV3/6, ZBTB18, GLI3) down; their target FOS and the p53–RB1–KAT5 checkpoint up

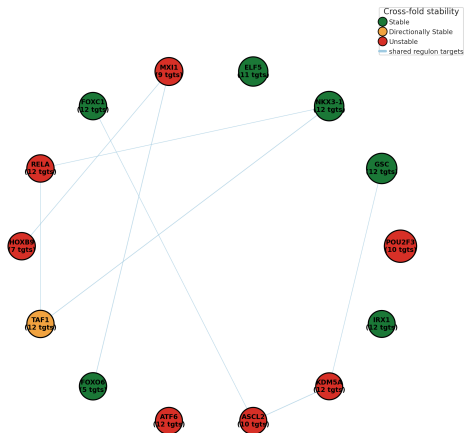
The preinvasive residual is a circuit, not a list



the same signature over all ~ 300 SCENIC regulons: FOS most context-elevated, IRF9/ETV6/SMAD4 most context-removed, and the stable set (dark) is what survives all five folds

The preinvasive residual is a circuit, not a list

R_{cond} -weighted CollecTRI regulon graph
(top context-sensitive TFs; node size = $|R_{\text{cond}}|$, color = cross-fold stability; edges = shared-target overlap)



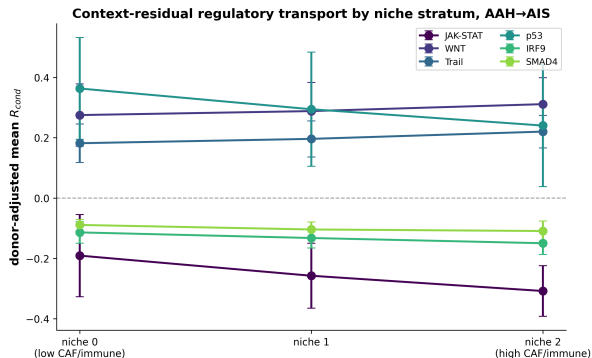
R_{cond} -weighted CollecTRI graph: node size = $|R_{\text{cond}}|$, colour = cross-fold stability, edges = shared targets

Same configuration. Different organ. Different method.

This work	Reyes et al., Cell 2026
human LUAD precursors context-residual transport	mouse PDAC, spontaneous p53 loss single-cell + spatial lineage
p53 ↑ RB1 ↑ SMAD4 ↓, with WNT / EGFR / AP-1 ↑	p53, CDKN2A, SMAD4 co-activated with oncogenic programs in progenitor- like cells

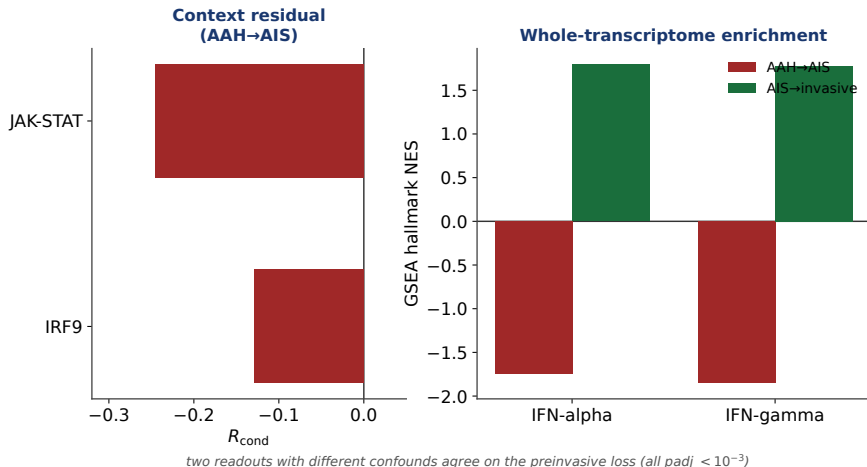
Including the same arm — SMAD4 — giving way.

The niche grades the contest



Less restraint, more drive at the inflamed end.

Interferon tone is lost *before* invasion



context residual and stage enrichment — different confounds — agree on the preinvasive loss

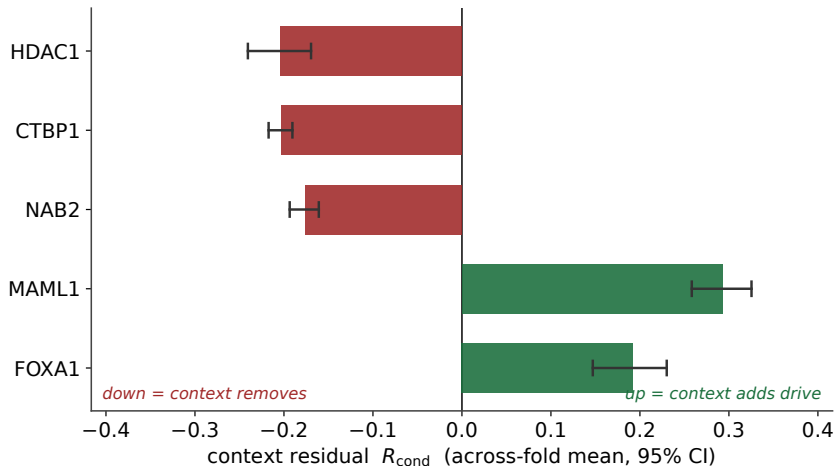
And in lesions with known outcomes, that loss predicts progression

Progressive lesions: **interferon lost**

Regressive lesions: **interferon retained**

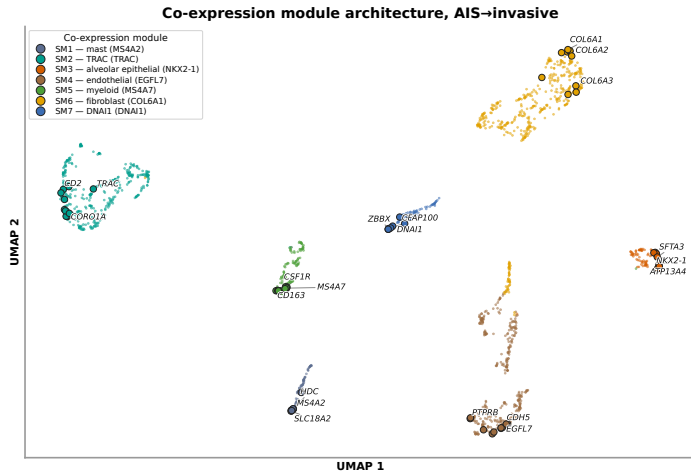
the only human premalignancy cohort profiled *and* followed to a known fate

Invasion is a change of architecture, not more drive



corepressors down, a coactivator and pioneer factor up — lineage identity is dismantled here, not before

Invasion is a change of architecture, not more drive



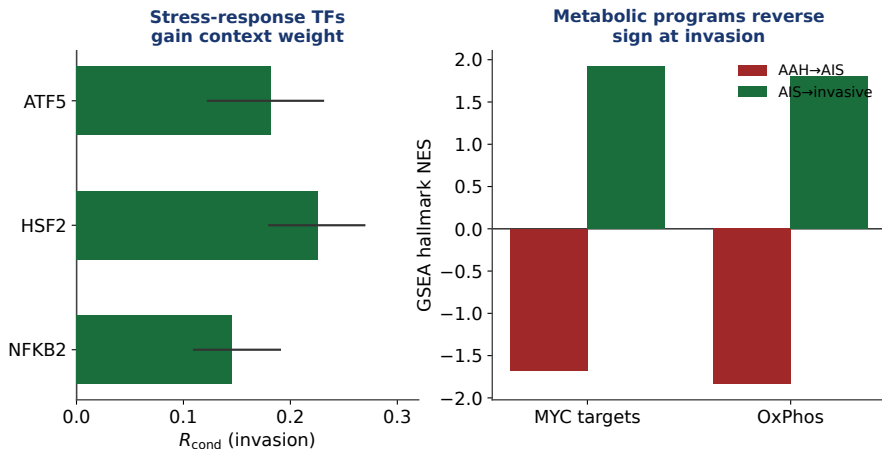
the actual co-expression architecture at this edge: seven modules, each anchored on a lineage hub gene

Invasion is a change of architecture, not more drive



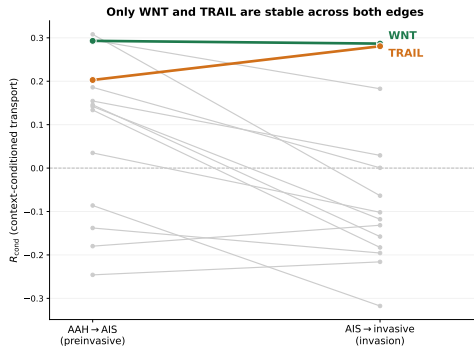
and the module that carries ciliary / axonemal identity (SM7) is the one whose enrichment collapses at invasion

An anabolic-stress module, on the same edge



anabolic demand and its stress-compensation machinery gain context weight together

Only two programs are invariant across the boundary



WNT TRAIL
everything else is edge-specific

VEGF goes *down* at invasion — and that is consistent

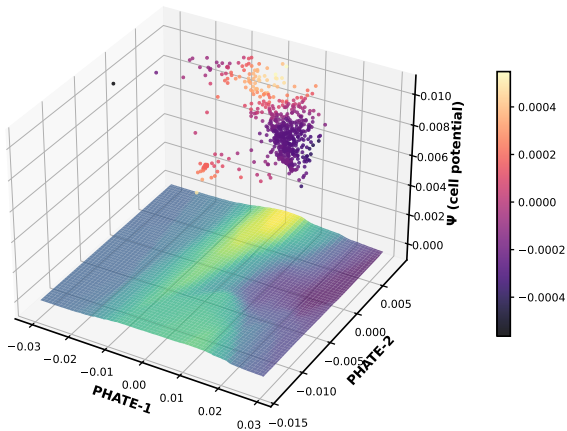
$$\text{VEGF} \quad R_{\text{cond}} = -0.318$$

largest stable pathway at the invasion boundary

A residual is not a level.

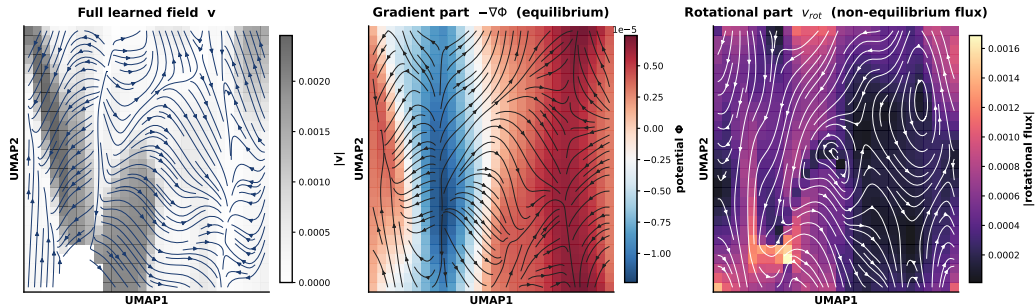
Progression is a directed flow, not a relaxation

Cell-potential landscape (AAH $\rightarrow$ AIS)



Progression is a directed flow, not a relaxation

Helmholtz-Hodge decomposition (full field): non-equilibrium flux fraction = 0.53



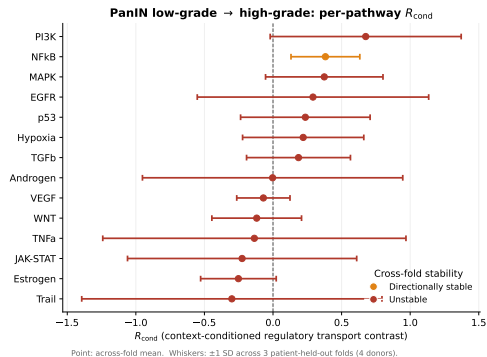
~1.3 million candidate signalling chains



two programs

macrophage APOE, PSAP, A2M → LRP1, EGFR → AP-1

The estimand transfers without over-calling



Four donors. **Zero stable programs.**

Act III

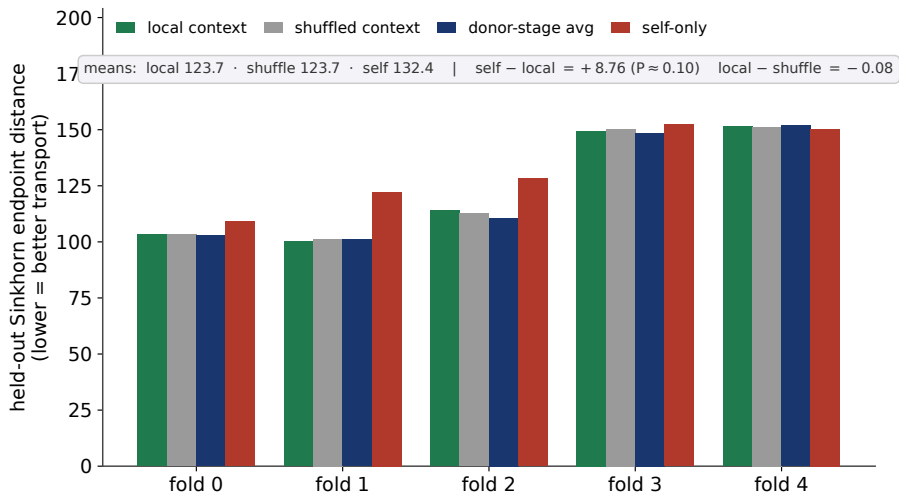
At what spatial scale does this hold?

The falsification ladder

Rung	What it removes
Local context	nothing — reference
Shuffled context	niche identity, distribution kept
Donor–stage average	within-specimen variation
Self-only	context entirely
Strict-state coupling	the ecological stratification

paired across five patient-held-out folds

The falsification ladder



A split verdict — and one half is a bound

Local over
shuffled context

$\Delta = +0.08$ $[-1.10, +1.26]$

Local over
self-only

$\Delta = +8.76$ $[-2.83, +20.35]$

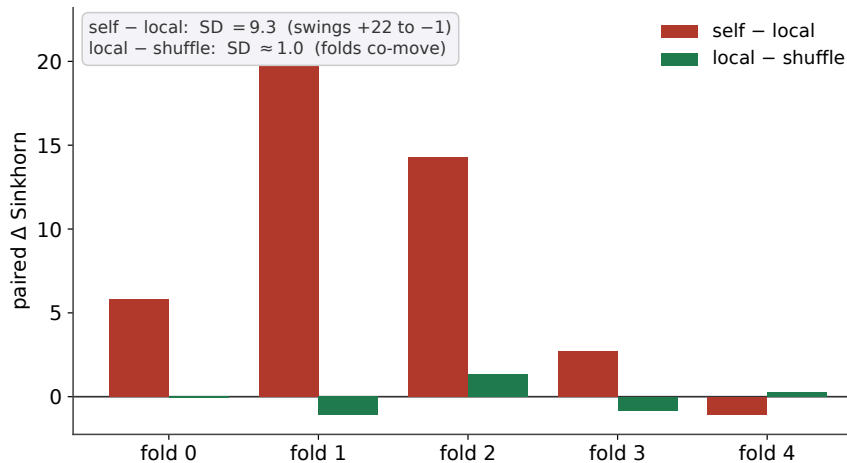
-10 0 10 20 30 40

paired Δ Sinkhorn (local – comparator), 95% CI across 5 folds

green = tight interval excluding a receiver-local benefit; grey = wide, underpowered, straddles zero

equivalence against shuffled context; underpowered against self-only ($P \approx 0.10$)

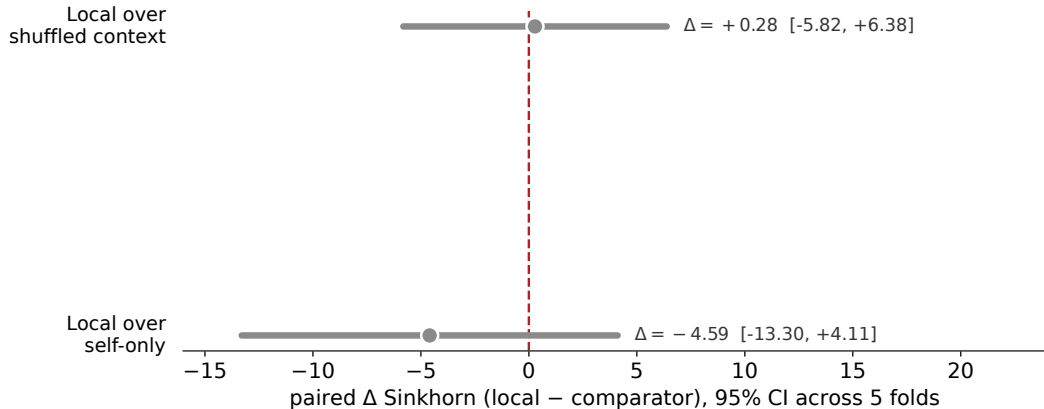
Why one interval is tight and the other is not



structural, not arbitrary: the shuffle keeps the same inputs in a different order, so its transport tracks local's fold for fold

the shuffle keeps the same input distribution, so its transport tracks local's fold for fold — self-only does not

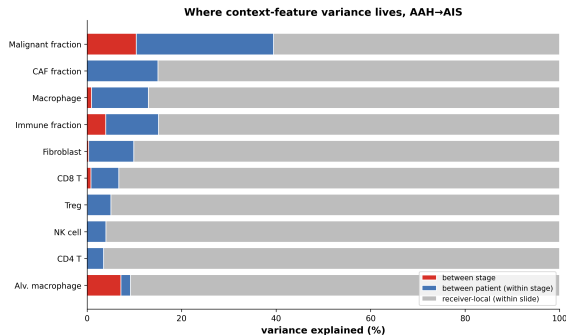
At the invasion boundary, both comparisons are null



both intervals straddle zero — zero shared donors on this edge

a reproducible per-program signature can coexist with no endpoint-predictive value — two different questions

Why: the ecology varies below our resolution



85–96% receiver-local and stage-invariant
between-patient 8–15% between-stage <1%

Immune hot and cold spots interchange over
tens of microns

Our spots are **55 μm** at **100 μm** centres

Two readings, and the experiment that separates them

A. Sub-resolution

local, but below 100 μm

better assays recover it

B. Specimen-scale

a field-level property

resolution will not help

Run the identical ladder at single-cell resolution.

A predicts recovery. B predicts continued equivalence.

Act IV

How the hypotheses resolve

	Hypothesis	Verdict
H ₁	Reproducible context residual	Met
H ₂	Receiver-local information vs shuffle vs self-only	Not met bounded equivalence underpowered
H ₃	Transfers across systems	Behaviourally

Weaker than the pre-registered alternative. I am reporting the weaker outcome.

- Cross-sectional — couplings are model alignments, not lineage maps
- **Two shared donors** on the primary edge; zero at invasion
- Spot resolution; ecology varies at or below it
- Context read at source and held fixed — no niche co-evolution
- 759 factors move through a **14-dimensional** gate
- Co-expression and communication are cross-method, not independent
- **Retrained-shuffle null: specified, not yet run**

1. Retrained-shuffle null — the outstanding confirmatory test
2. **The identical ladder at single-cell resolution**
3. Global/local decomposition, if ecology is field-scale
4. A mechanistic context channel: receptor occupancy, not a learned gate
5. Perturb APOE/PSAP/A2M $\rightarrow$ LRP1/EGFR $\rightarrow$ AP-1

Each specified precisely enough for someone else to run.

A falsifiable estimand.

A screen that rejected my own first design.

Reproducible, named biology on two stage edges.

A measured bound on where it stops.

Chris Bradburne

committee, collaborators, compute

Questions

Backup

B1 — Is 29/773 just the false-positive rate?

- $29/773 = 3.8\%$; a single 95% CI criterion has a 5% naive null rate. Fair question.
- Criterion is not one test: CI excluding zero **and** sign-consistency across five *disjoint patient* folds.
- Folds are correlated, so I do not claim an exact null rate.
- **The retrained-shuffle null is the right test and is not yet run.**
- Cheap intermediate: permute strata within donor, re-run the filter, report the null stable-count distribution.

B2 — The context channel is 14-dimensional

- Context enters only via $\mathbb{R}^{35} \rightarrow \mathbb{R}^{14} \rightarrow \mathbb{R}^{773}$.
- So coordination among transcription factors is partly *expected*, not evidential.
- What the bottleneck does *not* determine is which factors load or with what sign. A rank-restricted map does not force four repressors to oppose their target. **Sign structure is the informative part.**
- Pathway-level claims sit directly in the gated block and are strongest.

B3 — Per-fold ladder, preinvasive edge

	F0	F1	F2	F3	F4	mean
Local context	103.32	100.12	113.93	149.47	151.42	123.65
Self-only	109.14	122.16	128.19	152.20	150.36	132.41
self – local	+5.82	+22.04	+14.26	+2.73	–1.06	+8.76

SD 9.33, SEM 4.17, $t = 2.10$, $P \approx 0.10$. Against shuffle, $SD \approx 1.0$.

B4 — Decomposition credibility per fold

	F0	F1	F2	F3	F4	mean
D_ρ	0.949	0.963	1.008	0.984	0.922	0.965

- $D_\rho \leq 1$ means structured terms carry at least as much velocity as the neural residual — the interpretability precondition.
- **Fold 2 fails it.** Reported, not hidden.
- Which is why factor-level claims are stated below pathway-level ones.

B5 — Why is the held-out AUC *below* chance?

- Donor-held-out AUC for the self-only stage classifier: **0.419**, about 1.6 SE below 0.5.
- Not a bug — the signature of a **donor-entangled representation**. Cross-donor generalisation is anti-correlated, not absent.
- **Within-donor AUC is 0.833**, above chance in every shared donor.
- The representation is stage-discriminative; cross-patient transfer is what fails. Which is exactly why every estimate is patient-held-out.

- Linear endpoint baselines: 210.83 and 251.86 vs 123.65 fitted.
- Single-branch fields collapse: structured-only 2621.22, residual-only 654.81.
- Monolithic network fits *better* (121.59) — the price of an interpretable, program-resolved field. Stated, not omitted.
- Strict-state fits best (119.17) because the metric is scored in the space that coupling optimises. Near-circular, hence rejected on Tier 2.

- Conditional metric: Sinkhorn cost within held-out strata (≥ 5 per side), source-mass weighted, $\varepsilon = 0.05$.
- Primary because it evaluates the same conditioning that *defines* the estimand.
- Comparisons paired within fold, so estimator bias common across rungs cancels.
- An orthogonal metric — donor-held-out stage classification from predicted endpoints — would harden the negative. Named as future work.

- Four donors, ~ 100 graded receivers, two strata, three folds.
- Zero stable; 188 directionally stable.
- **True null and absence of power are not separable** without a positive control at matched n .
- The experiment that would earn the claim: inject a synthetic effect of known magnitude at $n = 4$ and show recovery.

- Every inline number traces to a versioned artifact; audit trail in the Ch3 header and Appendix C.15.
- Folds, hyperparameters, seeds: Appendix C.5–C.7.
- Per-fold ladders, 773-program stability atlas, transport diagnostics: Appendix C.8–C.11.
- Seeded per fold; training donors only; code and environment archived.

Any number on any slide can be traced live.

Tier	Analyses
Model-independent	stage DE, gene-set enrichment, stage discrimination
Residual-associated	ecological localisation, spatial autocorrelation
Residual-guided	SCENIC / hdWGCNA, gated communication

- Only the first tier is independent of the estimand.
- SCENIC overlap is **not enriched** beyond hypergeometric expectation ($0.95\text{--}1.14\times$, $P = 0.35\text{--}0.63$). Reported as such.

B11 — The ecology is spatially structured

- Moran's I , $k = 15$ per section, 55 sections, 787,816 spots. All 330 section-by-feature tests significant.
- Malignant clustering rises monotonically: $0.234 \rightarrow 0.458 \rightarrow 0.701$; Spearman $\rho = 0.673$.
- Getis–Ord hot-spot fraction nearly doubles.
- But macrophage autocorrelation is high and **stage-flat** ($0.443 \rightarrow 0.499 \rightarrow 0.467$).

Spatially organised within every specimen, only weakly graded by stage — the regime where local niche is real but hard to separate.

- Frozen before real data: recovery cosine ≥ 0.80 , seed agreement ≥ 0.90 , null gain < 0.10 , non-identifiable correlation < 0.50 .
- Five regimes including a confounded null and an alignment sweep.
- Conditional-stratified passed both tiers. Strict-state passed tier 1, failed tier 2. Progression-geometry failed both.
- Oracle pairing was the ceiling and is unavailable to any real analysis.

Two of three candidates rejected, including the one I started with.